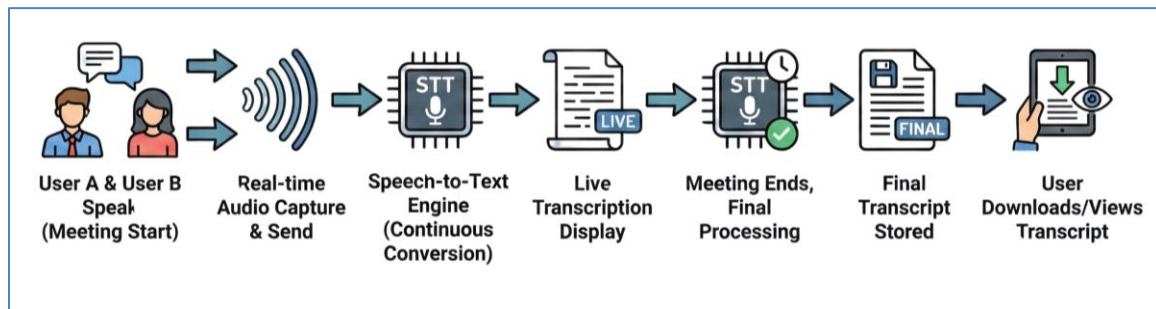


90 Days SDET Interview Prep

Day 6

Interview Question

“Google Meet has introduced a new transcription feature for meetings. How would you approach testing this feature?”



Step 1: Understand What the Feature Properly!

As a tester, I need to understand:

- What exactly the transcription feature does
- Expected accuracy levels (Setting up the metric)
- Supported languages and accents
- Operating conditions (noise, multiple speakers, long meetings)
- Any latency or availability expectations

What is Latency and Availability?

Latency:

- Time Delay between User Action and System Response (millisec /seconds)
- High latency means slow experience!
- Performance Testing or Latency Testing
 - Locust,k6 or Jmeter

Availability:

- Percentage of Time the System is up and accessible! (%)
- Low Availability Means System is not accessible!
- Reliability testing is Preferred
 - ChaosMonkey and Gremlin
 - <https://github.com/Netflix/chaosmonkey>
 - Enterprise tool: <https://www.gremlin.com/gremlin-free-software>

Where do we get this information from?

- Product Requirement Documents (PRD)
- Technical / Design Docs
- Discussions with Product Managers & Developers
- Internal documentation or release notes

Step 2: Define the Scope of Testing

Once we understand the feature, define what all we are going to test.

For transcription, the scope must include:

- ✓ Accuracy Testing
- ✓ Language & Accent Coverage
- ✓ Performance & Latency
- ✓ Robustness (noise, interruptions, overlapping speech)
- ✓ Privacy & Compliance (PII, data handling)

In this discussion, we deep-dive into Accuracy Testing first.

STT Testing:

- ✓ The *therapist* said he saw *the rapist* today [Tiny Pause]
- ✓ We need to deliver two too many features at 2:02
- ✓ Let's eat grandma, not later [Punctuation]
- ✓ *He knows* no one knows his *nose* [Same Sounds Different Meaning]
- ✓ I said Friday—sorry—Thursday—no wait—Wednesday.
- ✓ So the deploy— **cough** —failed again [Understanding Noise]

Step 3: Define the Key Accuracy Metric

Accuracy cannot be subjective. [It needs to be measured well]

Define a clear, industry-standard metric:

Word Error Rate (WER)

WER measures:

- Substitutions: A wrong word replaces the correct word
 - Reference: “*I love automation testing*”
 - Transcript: “*I love **automation texting***”
- Insertions: Extra word added that was never spoken
 - Reference: “Deploy the build”
 - Transcript: “Deploy the new build”
- Deletions: A word that was spoken is missing
 - Reference: “Start the meeting recording”
 - Transcript: “Start the meeting”

Compared against a reference text.

Lower WER = better transcription accuracy.

Step 4: How Do We Test Accuracy?

Manual Way (Not Scalable)

- Speak in a meeting
- Look at the transcript
- Visually check correctness

It's not reliable or scalable. (Good For Sanity)

Step 5: Automation-Based Accuracy Testing (Scalable Approach)

Idea: We use predefined reference text and automation tools to measure accuracy objectively.

Reference Text Strategy (Very Important)

The reference text is pre-decided.

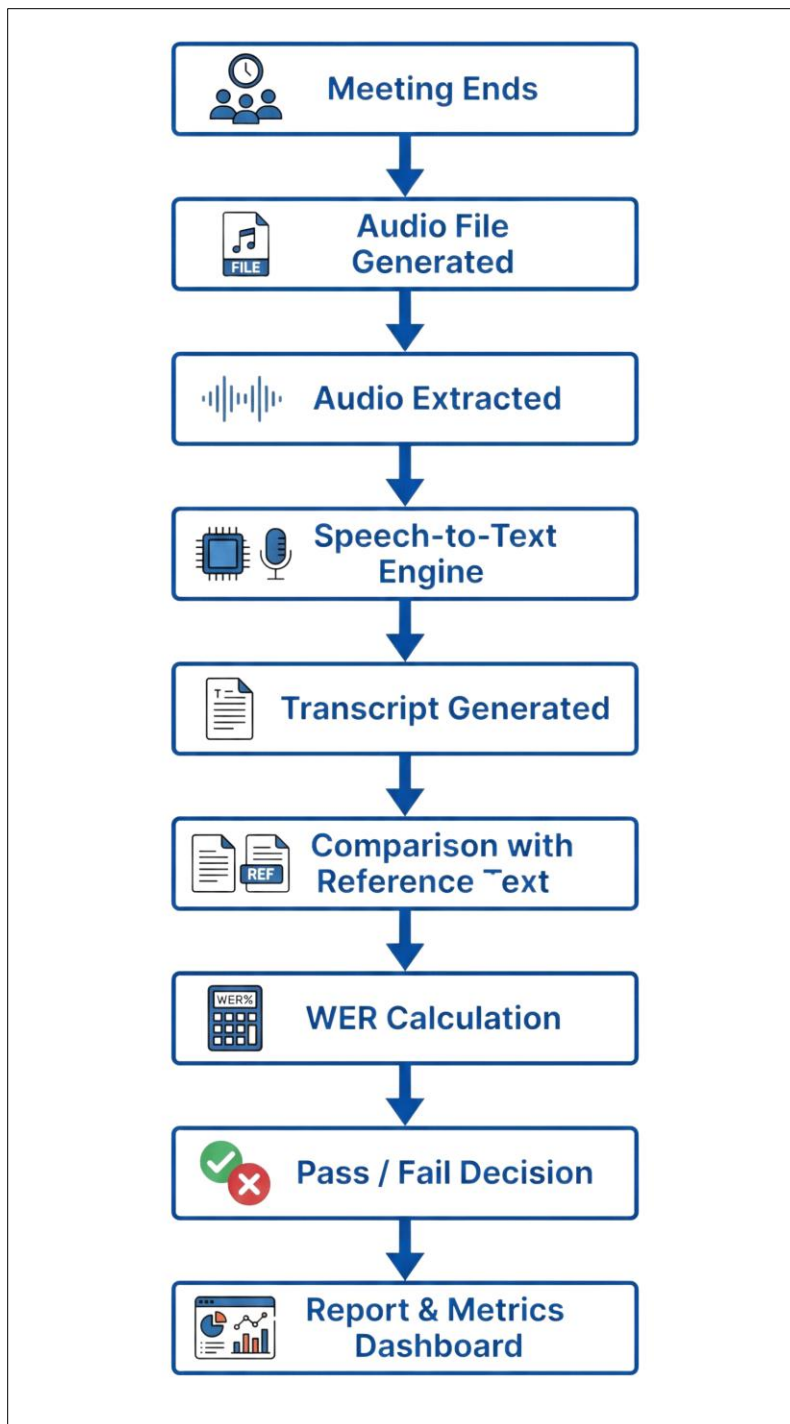
- *LibriSpeech: <https://www.openslr.org/12>*

How it works

- We prepare a known sentence or paragraph
- The speaker reads exactly that content
- The system generates a transcript
- We compare:
- Reference Text vs Generated Transcript

This gives us clean, repeatable, measurable results.

End-to-End Accuracy Testing Flow



Step 6: Automation Libraries & How They Fit in the Flow

Tool / Library	Where It Fits	What It Does
SCTK	Post-transcription	Calculates Word Error Rate (WER)
Levenshtein Distance	Post-transcription	Measures text similarity
NLTK / NLP libs	Text processing	Cleaning, tokenization
PyTest	Test execution	Automates test runs
CI Pipeline	Continuous testing	Prevents accuracy regression

And For Java

Purpose	Java Library
Levenshtein Distance	Apache Commons Text
NLP Processing	Stanford NLP / OpenNLP
Test Framework	JUnit / TestNG
Automation	Maven / Gradle + CI

Automated STT Quality Pipeline

